

Experiment 03:

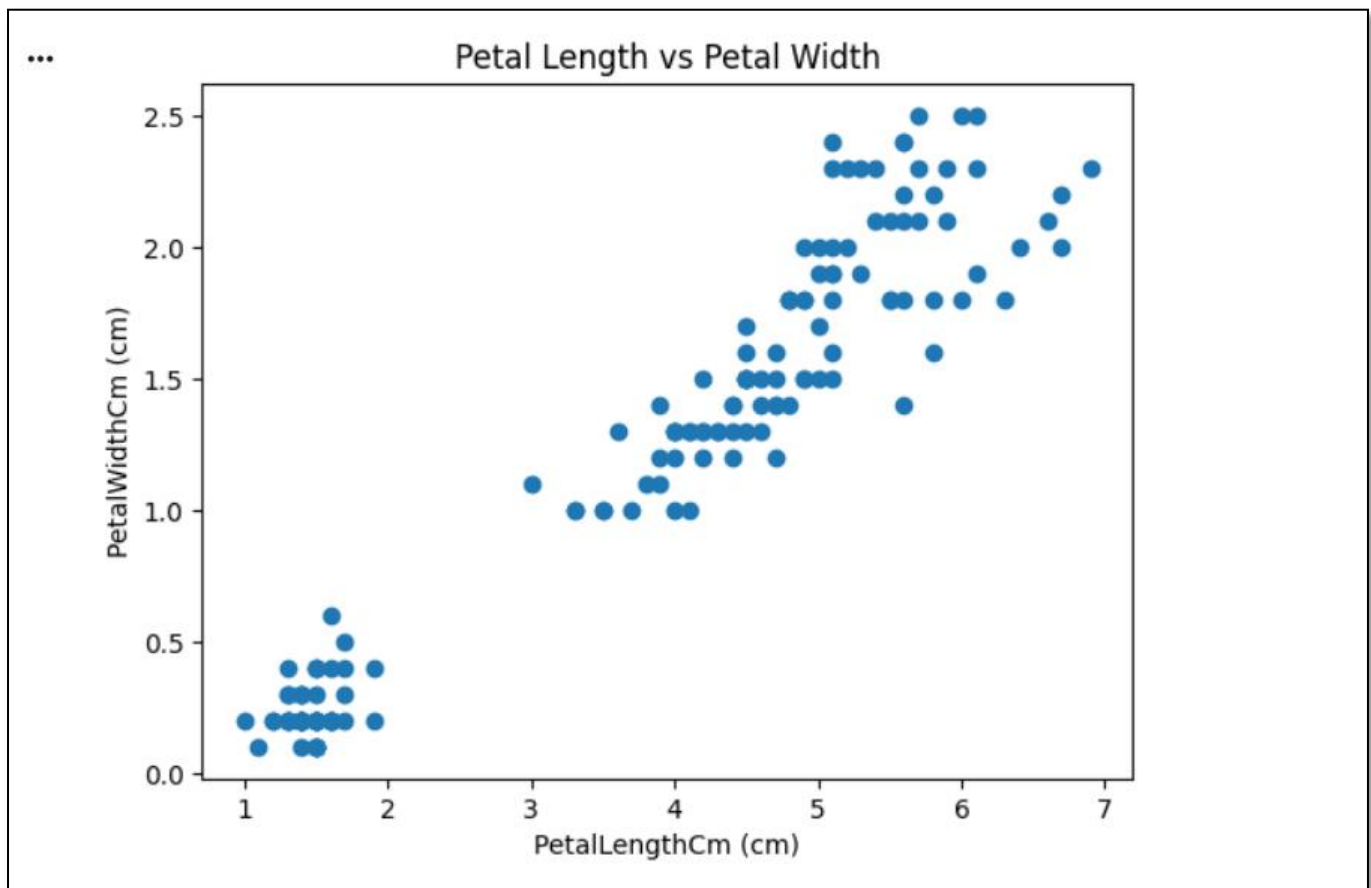
Laboratory Exercise

1. Scatterplot

Code:

```
plt.scatter(df['PetalLengthCm'], df['PetalWidthCm'])  
plt.title('Petal Length vs Petal Width')  
plt.xlabel('PetalLengthCm (cm)')  
plt.ylabel('PetalWidthCm (cm)')  
plt.show()
```

Output:

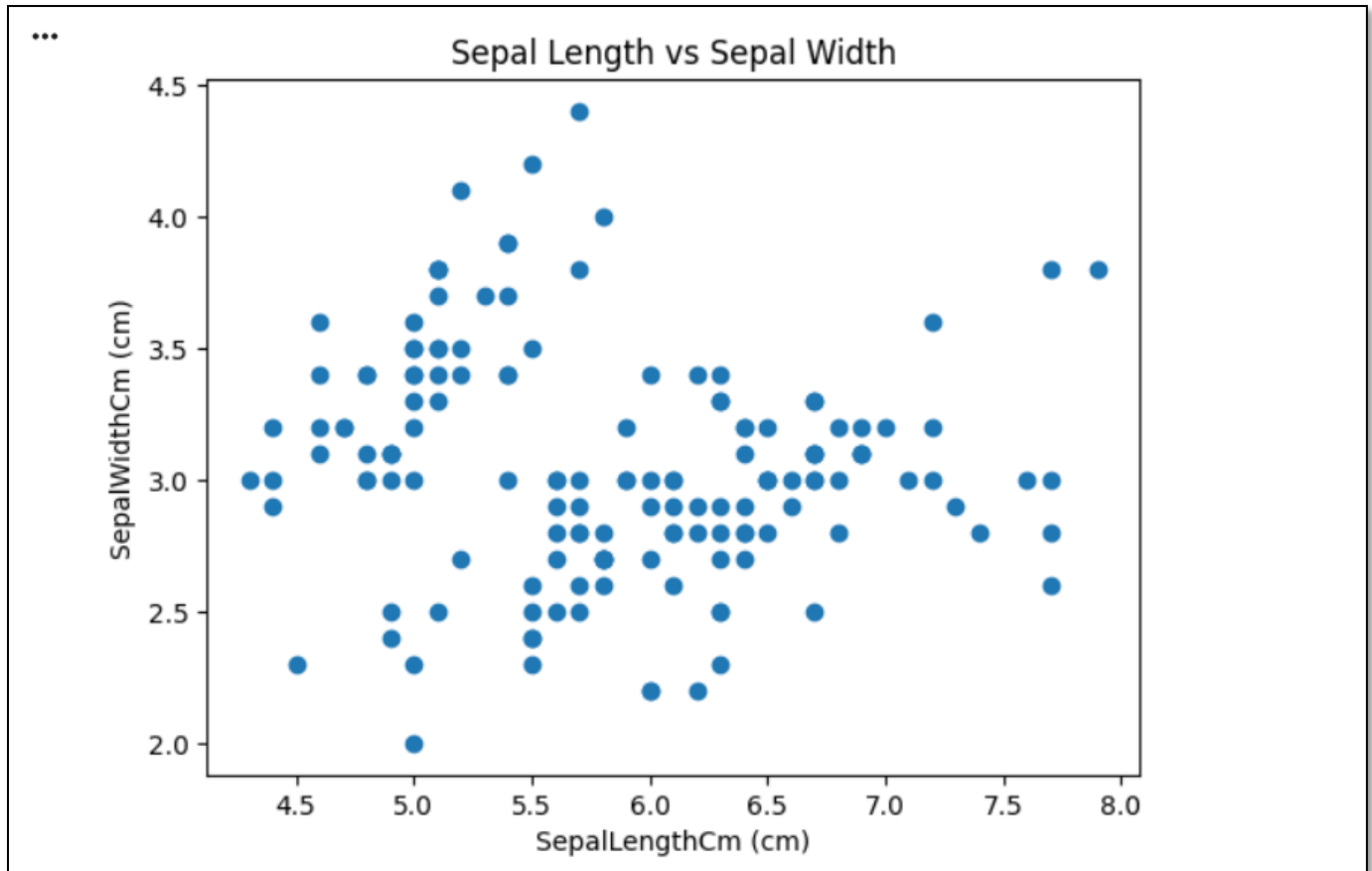


Observation:

- As petal length increases, petal width also increases.
- The points follow an upward trend.
- This indicates a strong positive relationship.
- The data is not random.

Code:

```
plt.scatter(df['SepalLengthCm'], df['SepalWidthCm'])  
plt.title('Sepal Length vs Sepal Width')  
plt.xlabel('SepalLengthCm (cm)')  
plt.ylabel('SepalWidthCm (cm)')  
plt.show()
```

Output:**Observation:**

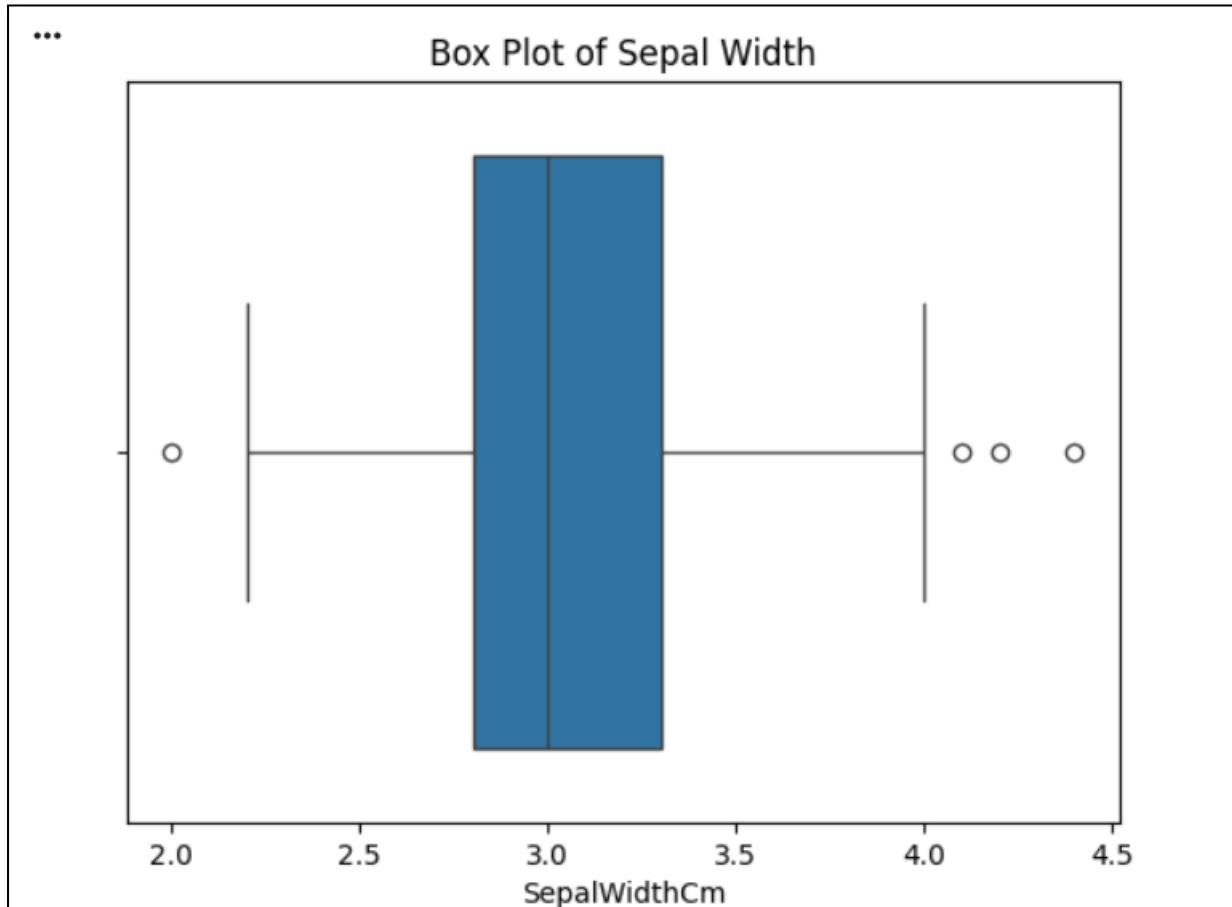
- Sepal length and sepal width show a weak negative relationship.
- The points are more scattered compared to petal measurements.
- No strong linear trend is observed.

2. Boxplot

Code:

```
sns.boxplot(x=df['SepalWidthCm'])  
plt.title("Box Plot of Sepal Width")  
plt.show()
```

Output:

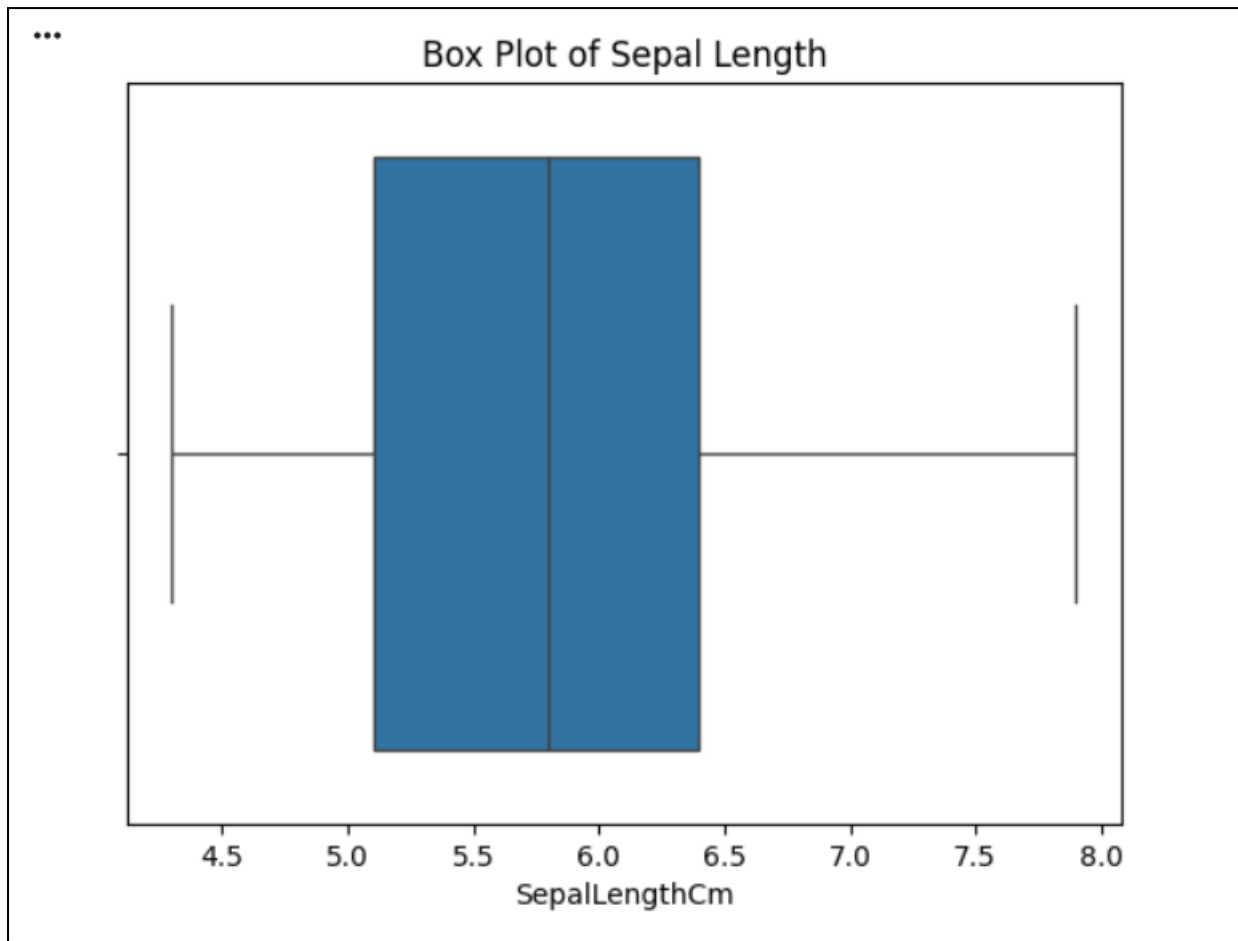


Observation:

- The median is around **3.0 cm**.
- The middle 50% of values lie approximately between **2.8 cm and 3.3 cm**.
- The whiskers extend from about **2.3 cm to 4.0 cm**.
- A few outliers are present.

Code:

```
sns.boxplot(x=df['SepalLengthCm'])  
plt.title("Box Plot of Sepal Length")  
plt.show()
```

Output:**Observation:**

- The median sepal length is around **5.8 cm**.
- Most values lie between **5.1 cm and 6.4 cm** (middle 50% of the data).
- The whiskers show the overall spread of the data.
- **No significant outliers are observed.**
- The distribution appears fairly balanced.

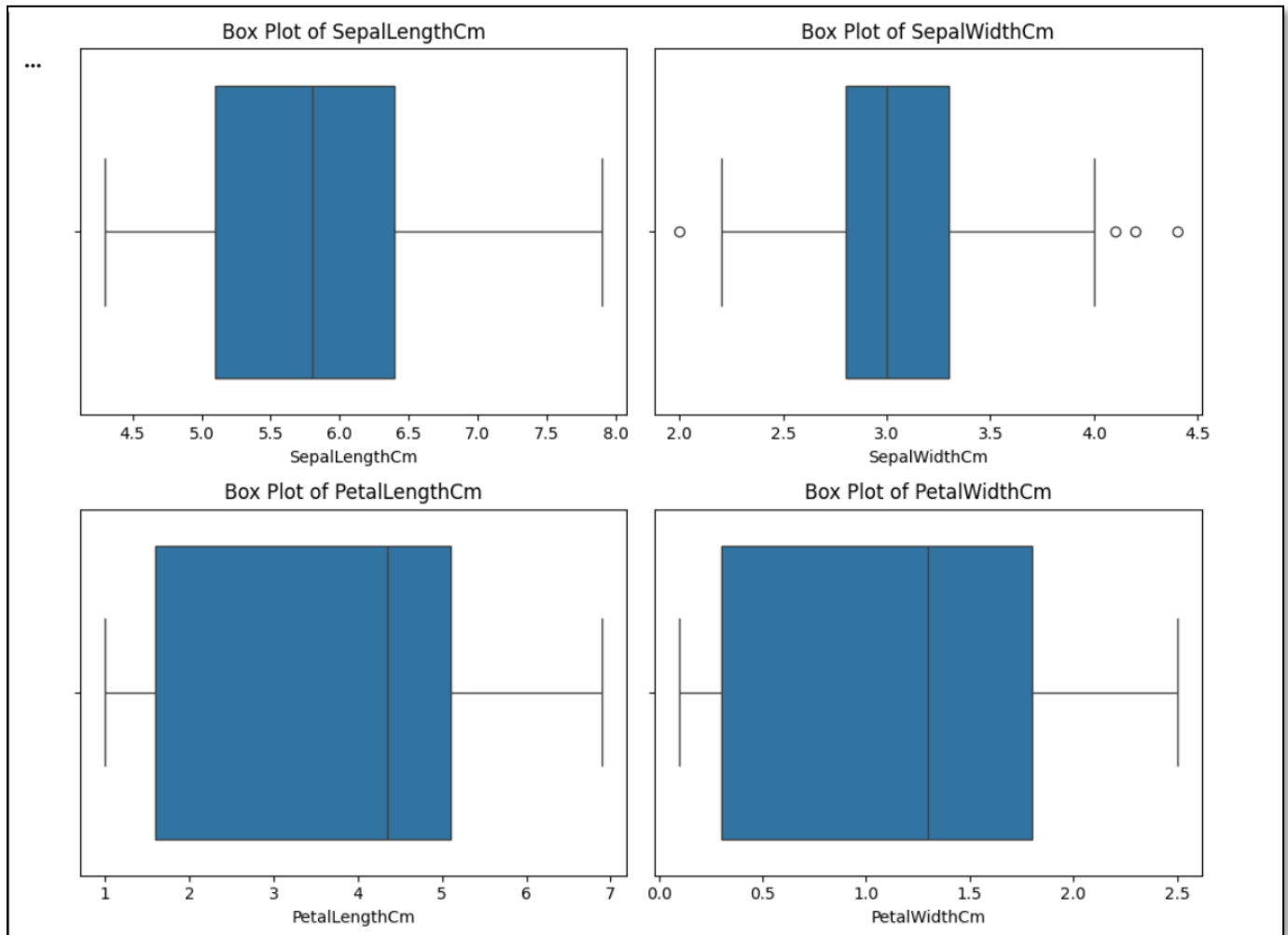
Code:

```
fig, axes = plt.subplots(2, 2, figsize=(10, 8))

columns = ["SepalLengthCm", "SepalWidthCm", "PetalLengthCm", "PetalWidthCm"]

for ax, col in zip(axes.flat, columns):
    sns.boxplot(x=df[col], ax=ax)
    ax.set_title(f"Box Plot of {col}")

plt.tight_layout()
plt.show()
```

Output:**Observation:**

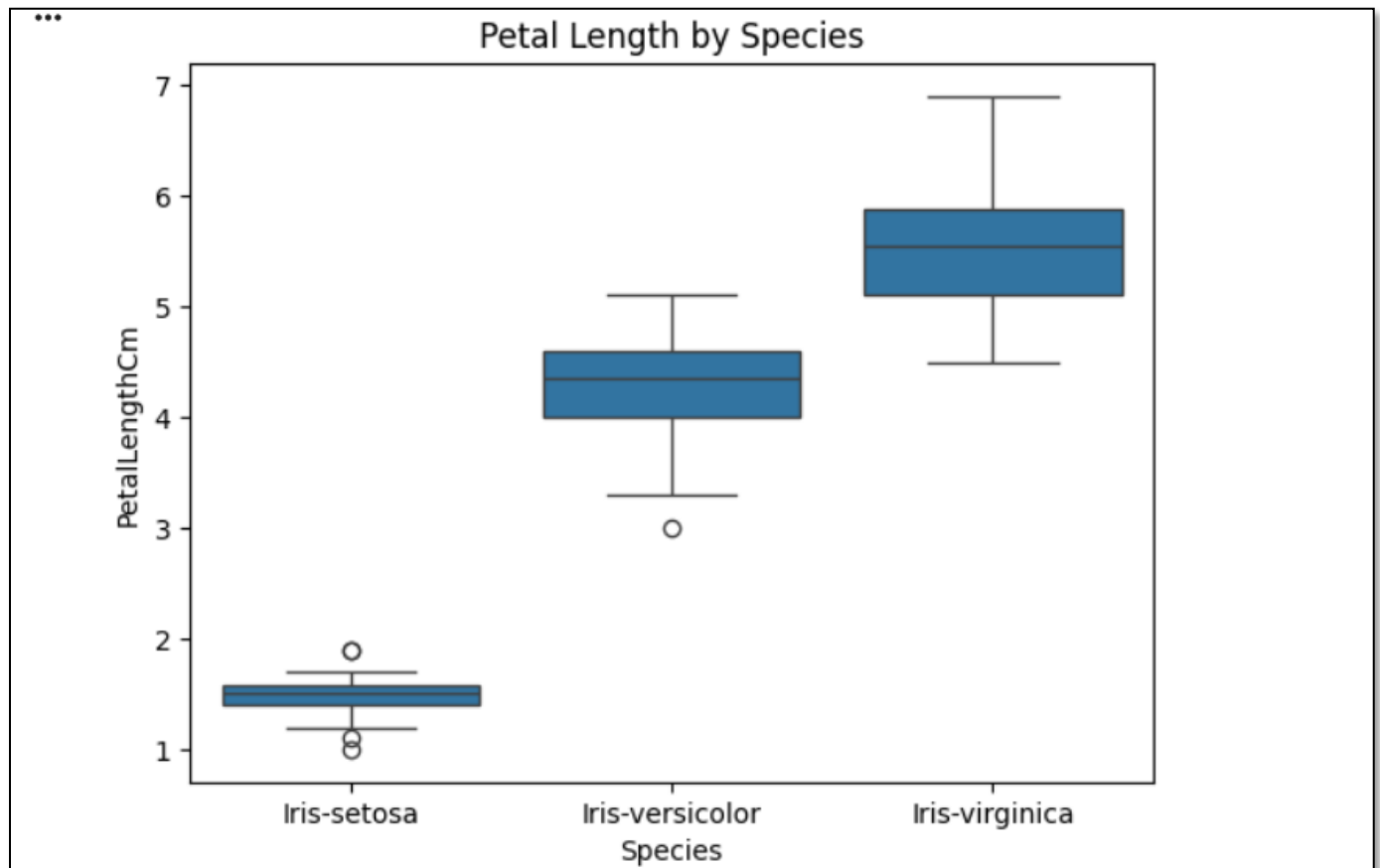
- The four box plots show the distribution of all numerical attributes.
- Each plot displays the median, spread, and overall distribution.
- Outliers, if present, can be identified easily.
- Petal measurements have a wider spread than sepal measurements.

Code:

```
sns.boxplot(data=df, x='Species', y='PetalLengthCm')  
plt.title('Petal Length by Species')  
plt.show()
```

Observation:

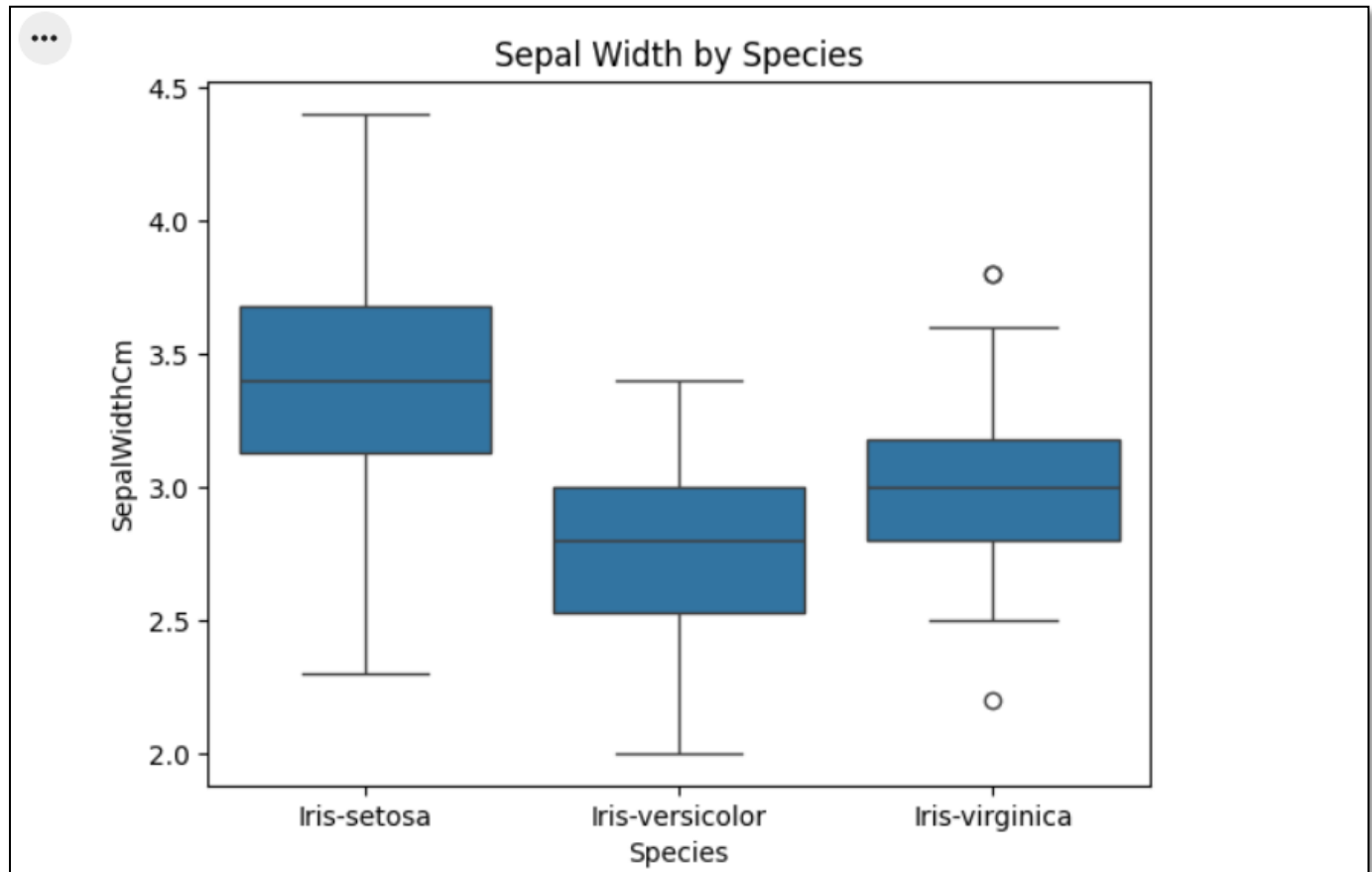
- Petal length varies significantly among the three species.
- Iris-setosa has the smallest petal length.
- Iris-virginica has the largest petal length.
- The species are clearly separated.

Output:**Code:**

```
sns.boxplot(data=df, x='Species', y='SepalWidthCm')  
plt.title('Sepal Width by Species')  
plt.show()
```

Observation:

- Sepal width differs across species.
- **Iris-setosa** generally has higher sepal width values.
- **Versicolor** and **Virginica** have overlapping distributions.
- Only a small variation is observed compared to petal length.

Output:

3. Countplot

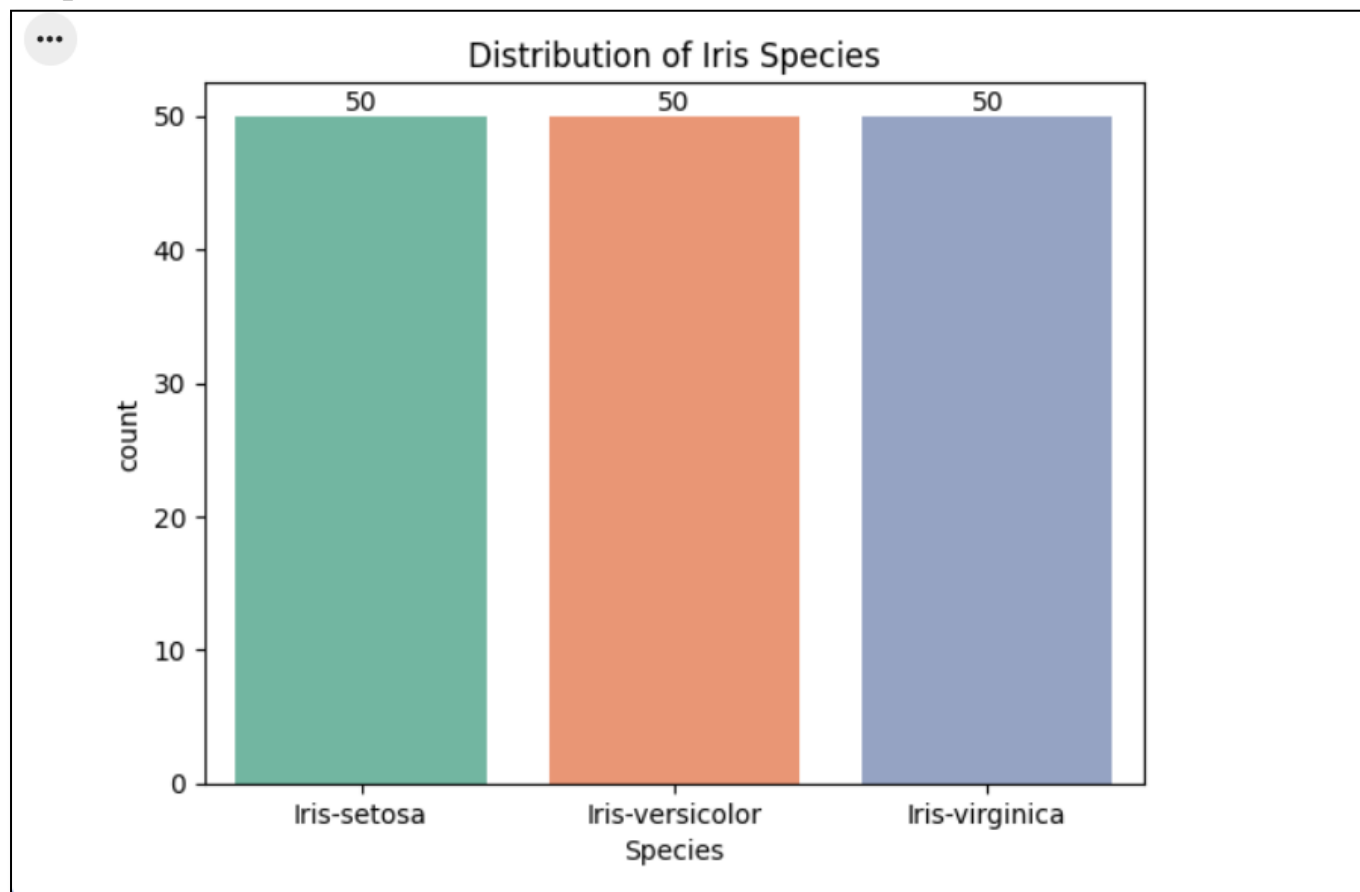
Code:

```
ax = sns.countplot(
    data=df,
    x='Species',
    hue='Species',
    palette='Set2',
    legend=False
)
# Display count above each bar
for container in ax.containers:
    ax.bar_label(container)

plt.title('Distribution of Iris Species')
plt.show()
```

Observation:

- Each species has **50 observations**.
- The dataset is balanced.
- All three species have equal representation.

Output:**4. Heatmap****Code:**

```
correlation = df.corr(numeric_only=True)
correlation
```

Output:

A heatmap showing the correlation matrix for the Iris dataset. The table has 6 columns: Id, SepalLengthCm, SepalWidthCm, PetalLengthCm, and PetalWidthCm. The diagonal elements are all 1.000000. The off-diagonal elements show the correlation between the features.

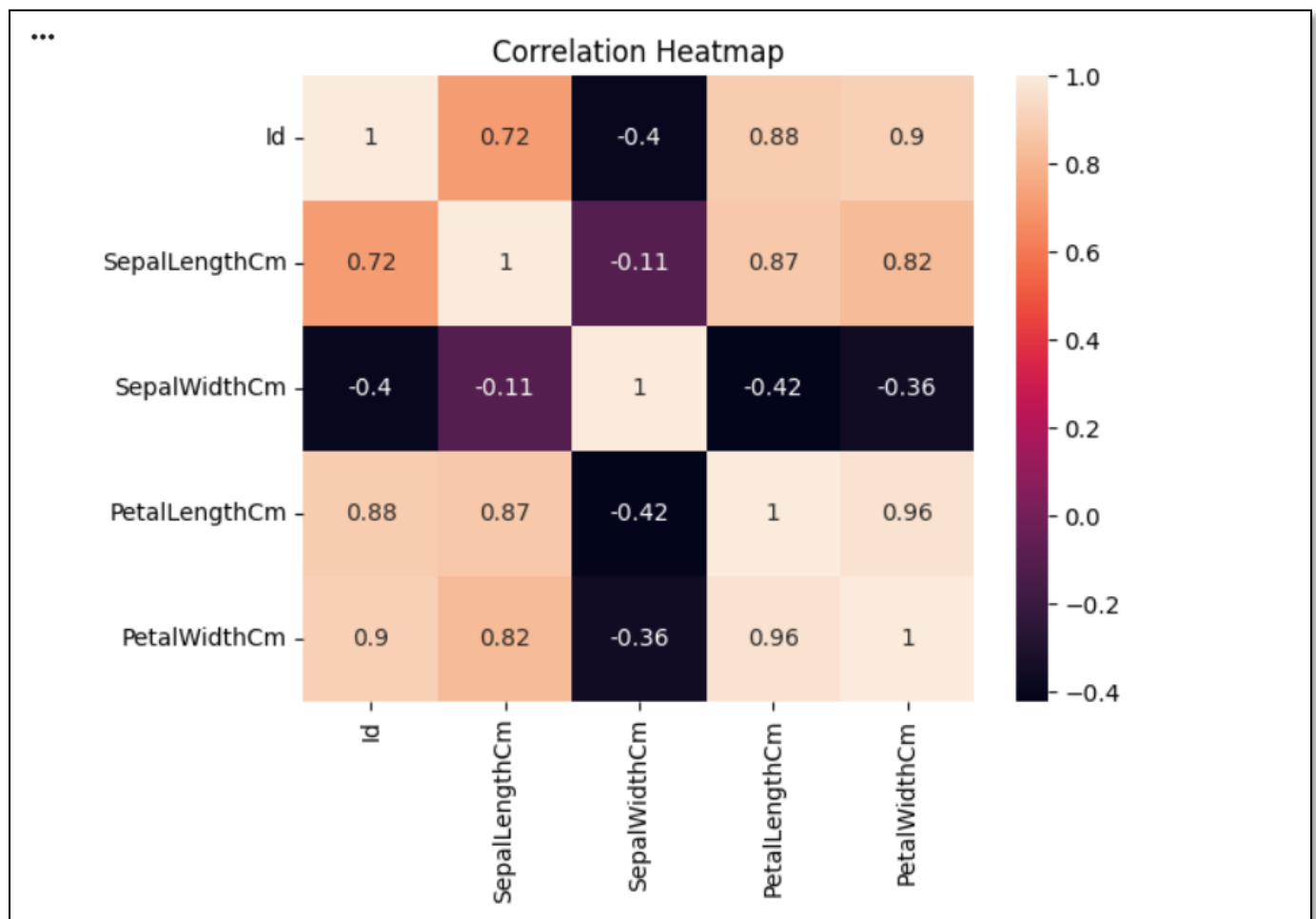
	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
Id	1.000000	0.716676	-0.397729	0.882747	0.899759
SepalLengthCm	0.716676	1.000000	-0.109369	0.871754	0.817954
SepalWidthCm	-0.397729	-0.109369	1.000000	-0.420516	-0.356544
PetalLengthCm	0.882747	0.871754	-0.420516	1.000000	0.962757
PetalWidthCm	0.899759	0.817954	-0.356544	0.962757	1.000000

Observation:

- The correlation matrix displays the relationship between numerical attributes.
- Values close to 1 indicate a strong positive correlation.
- Values close to 0 indicate a weak relationship.
- This matrix is used to create the heatmap in the next step.

Code:

```
sns.heatmap(correlation, annot=True)
plt.title('Correlation Heatmap')
plt.show()
```

Output:**Observation:**

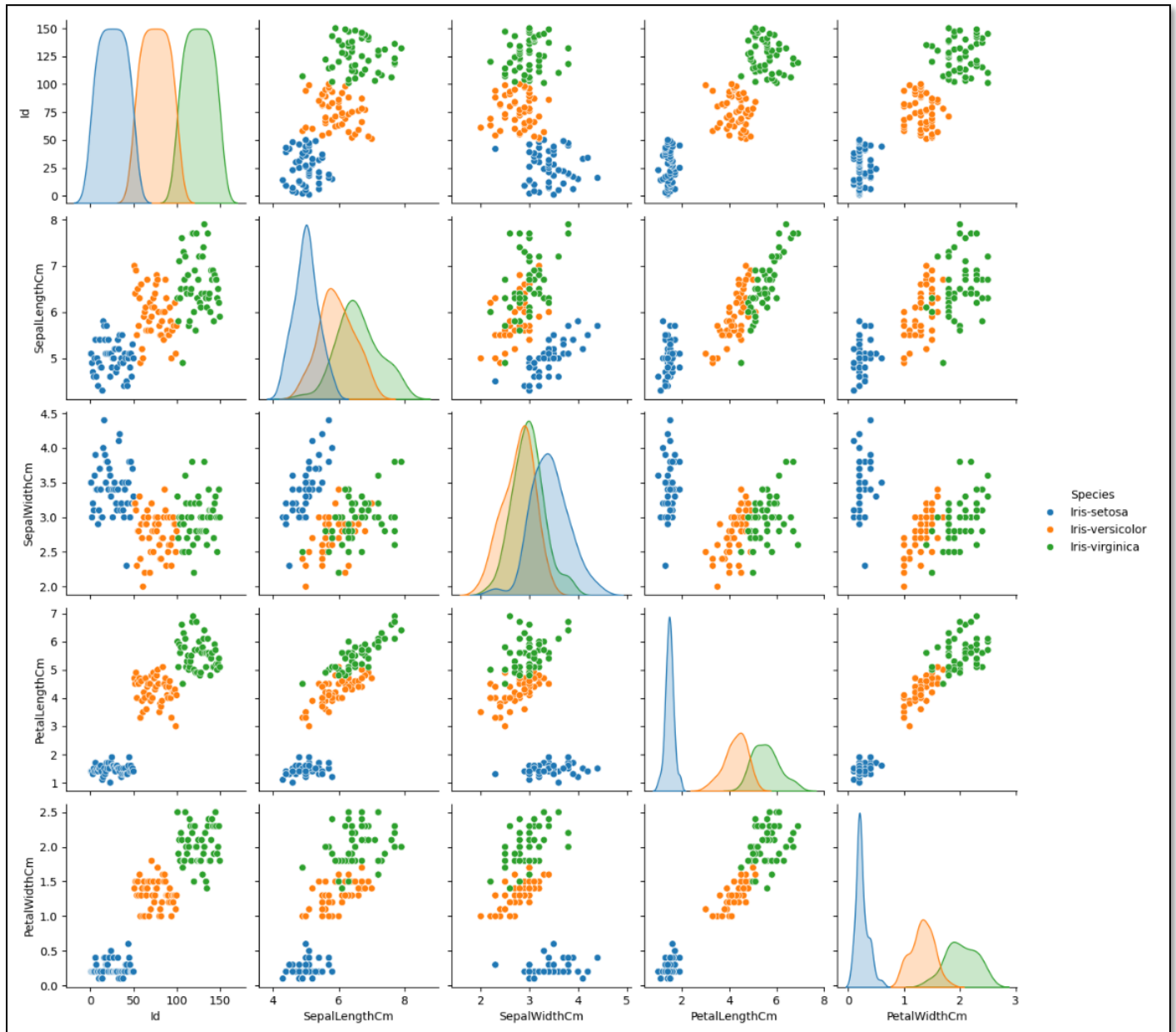
- Petal length and petal width have a **strong positive correlation**.
- Sepal length also shows a positive correlation with petal measurements.
- Sepal width has a comparatively weaker correlation with the other features.
- The heatmap makes it easy to identify relationships between numerical attributes.

5. Pairplot

Code:

```
sns.pairplot(df, hue='Species')  
plt.show()
```

Output:



Observation:

- The pair plot compares all numerical features simultaneously.
- Petal Length and Petal Width show a strong positive relationship.
- Iris-setosa is clearly separated from the other two species.
- Versicolor and Virginica overlap slightly in some plots.

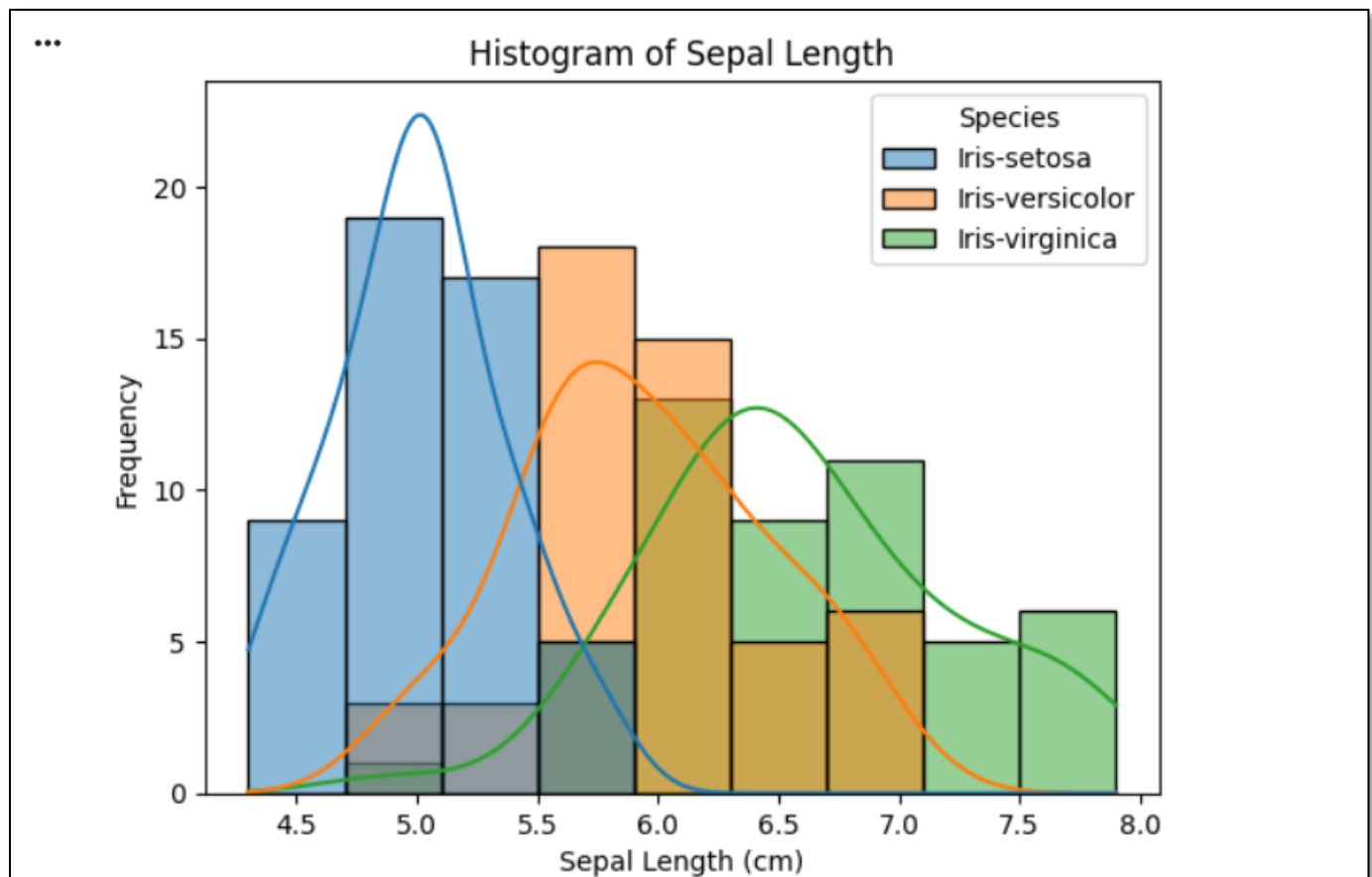
6. Histogram

Code:

```
sns.histplot(data=df, x='SepalLengthCm', hue='Species', kde=True)
```

```
plt.title("Histogram of Sepal Length")  
plt.xlabel("Sepal Length (cm)")  
plt.ylabel("Frequency")  
plt.show()
```

Output:



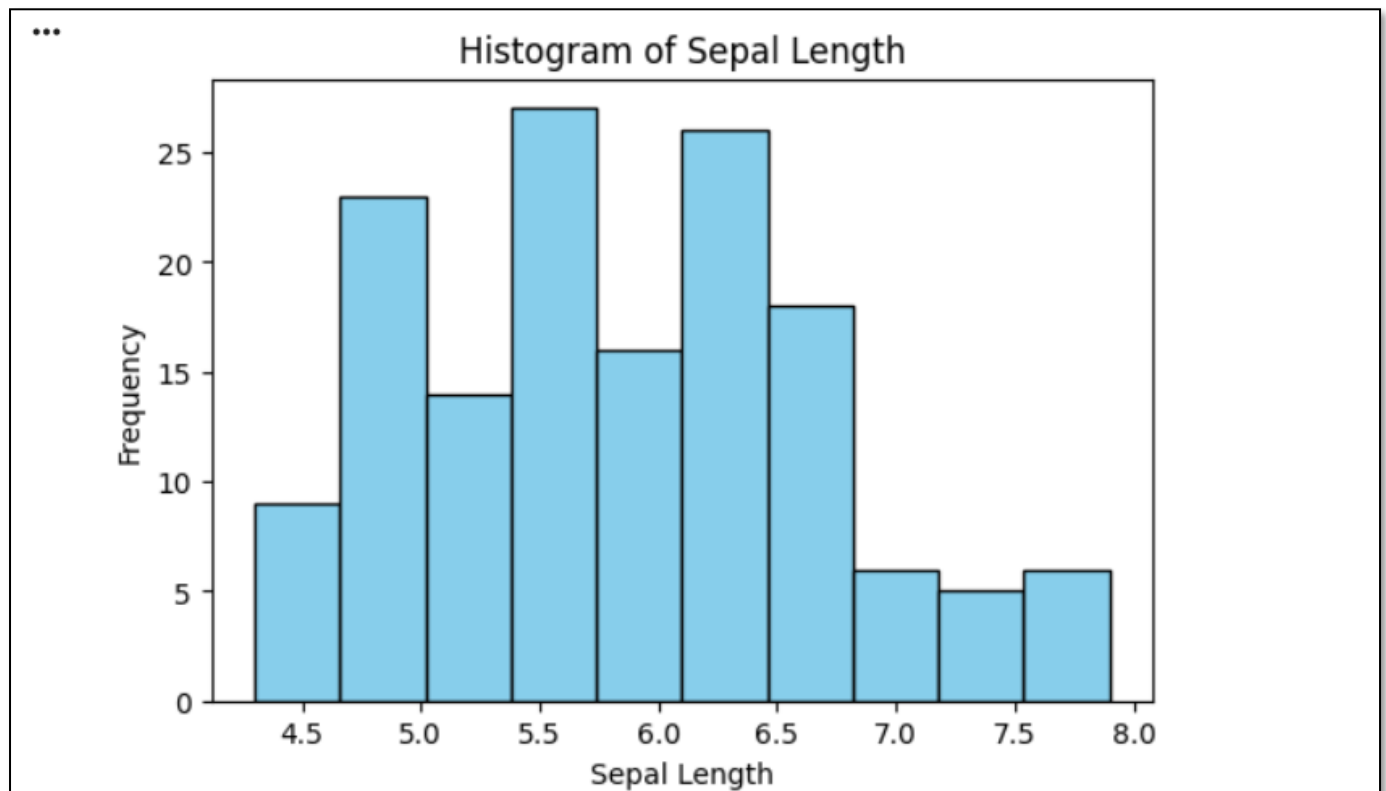
Observation:

- The histogram shows the distribution of Sepal Length.
- Iris-setosa generally has smaller sepal lengths.
- Versicolor and Virginica overlap in some regions.
- The KDE curve helps visualize the distribution smoothly.

Code:

```
plt.figure(figsize=(6,4))
plt.hist(x=df['SepalLengthCm'], bins=10, color='skyblue', edgecolor='black')

plt.title('Histogram of Sepal Length')
plt.xlabel('Sepal Length')
plt.ylabel('Frequency')
plt.show()
```

Output:**Observation:**

- The histogram displays the frequency distribution of sepal length.
- Most observations are concentrated in the middle range.
- The data appears fairly continuous with no major gaps.
- The distribution is slightly asymmetric.

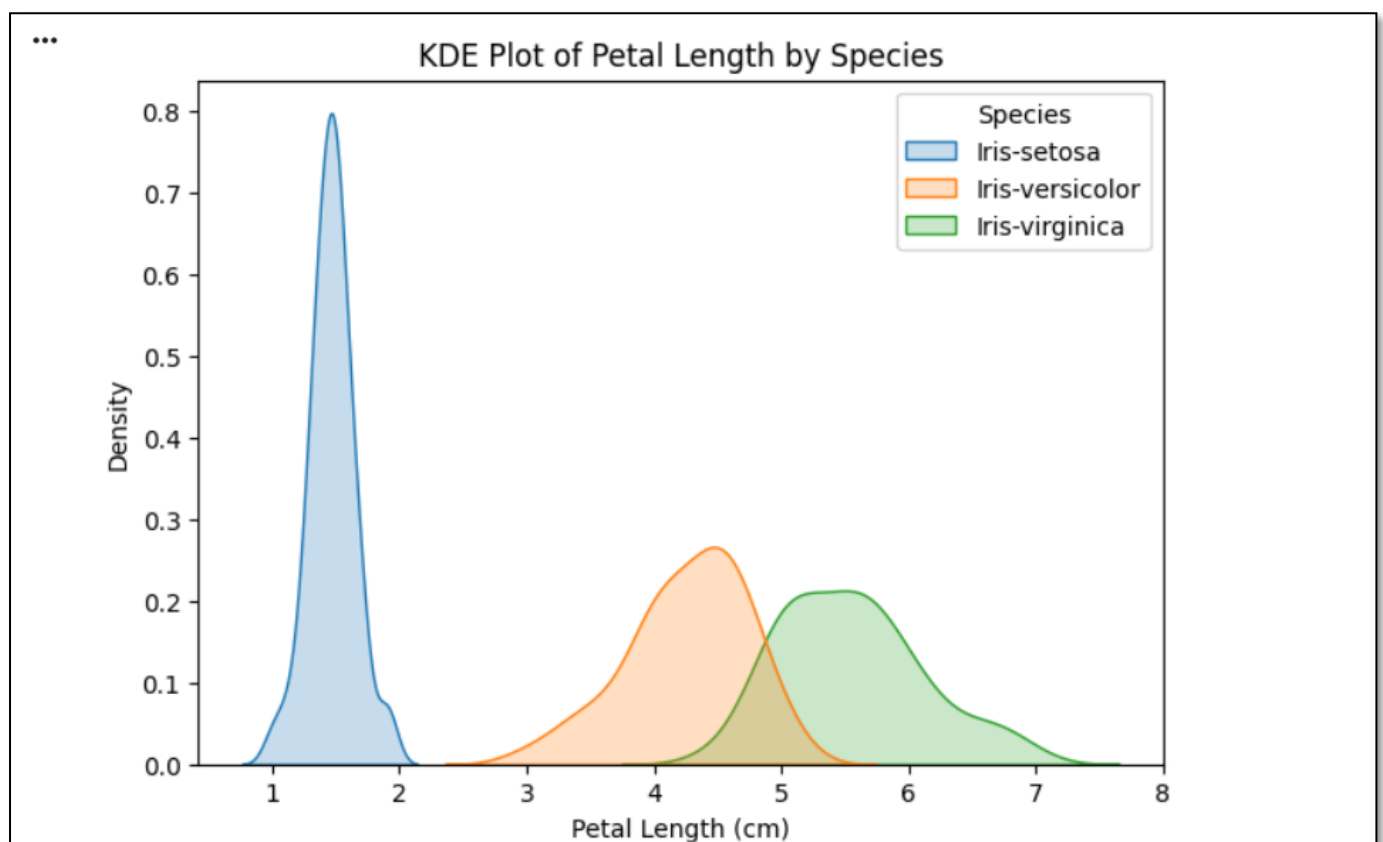
7. KDE-plot

Code:

```
plt.figure(figsize=(7,5))
sns.kdeplot(data=df, x='PetalLengthCm', hue='Species', fill=True)

plt.title('KDE Plot of Petal Length by Species')
plt.xlabel('Petal Length (cm)')
plt.ylabel('Density')
plt.show()
```

Output:



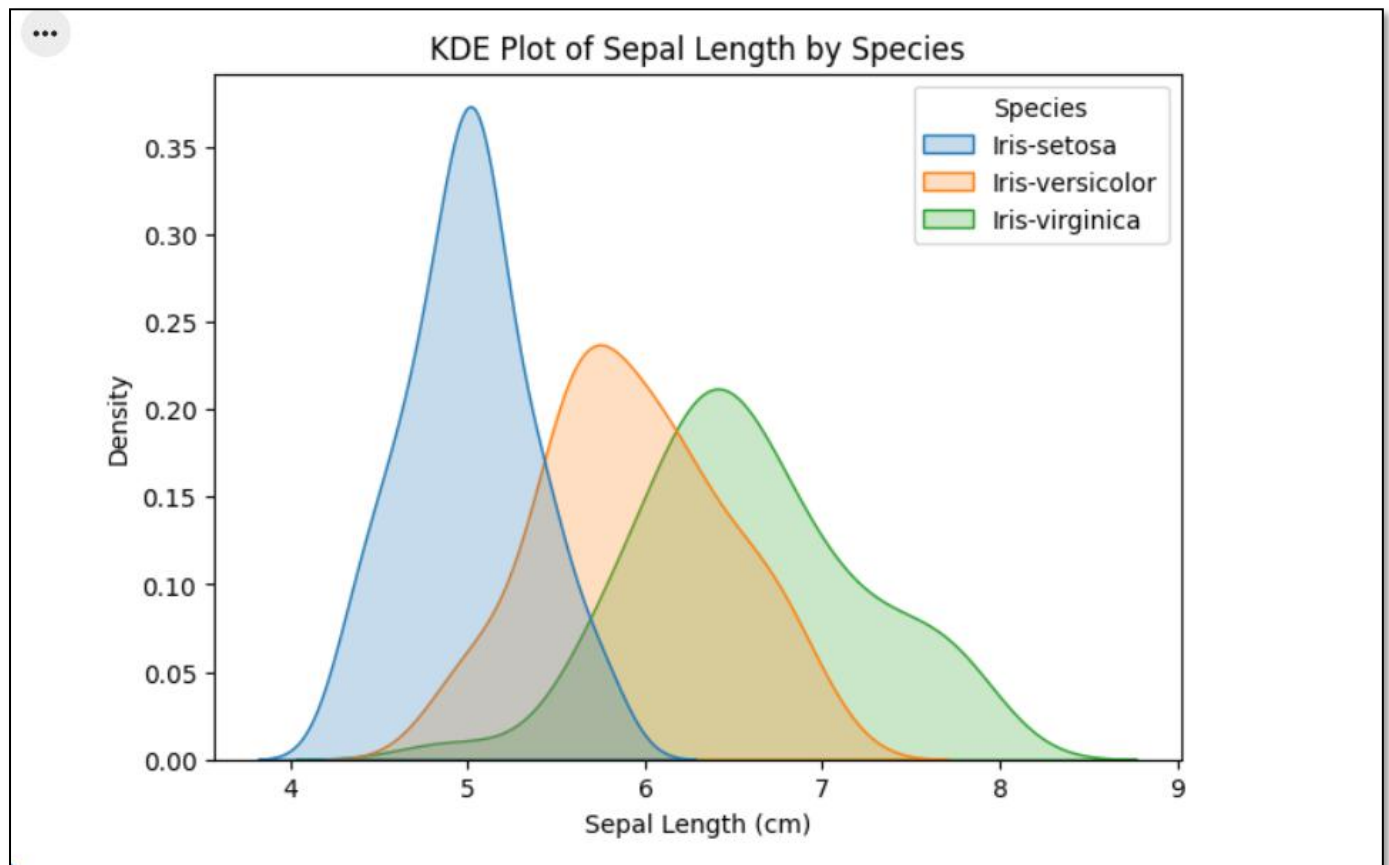
Observation:

- The KDE plot shows the density distribution of petal length.
- Iris-setosa has the smallest petal lengths.
- Virginica has the highest petal lengths.
- The three species form distinct density curves with slight overlap between Versicolor and Virginica.

Code:

```
plt.figure(figsize=(7,5))
sns.kdeplot(data=df, x='SepalLengthCm', hue='Species', fill=True)

plt.title('KDE Plot of Sepal Length by Species')
plt.xlabel('Sepal Length (cm)')
plt.ylabel('Density')
plt.show()
```

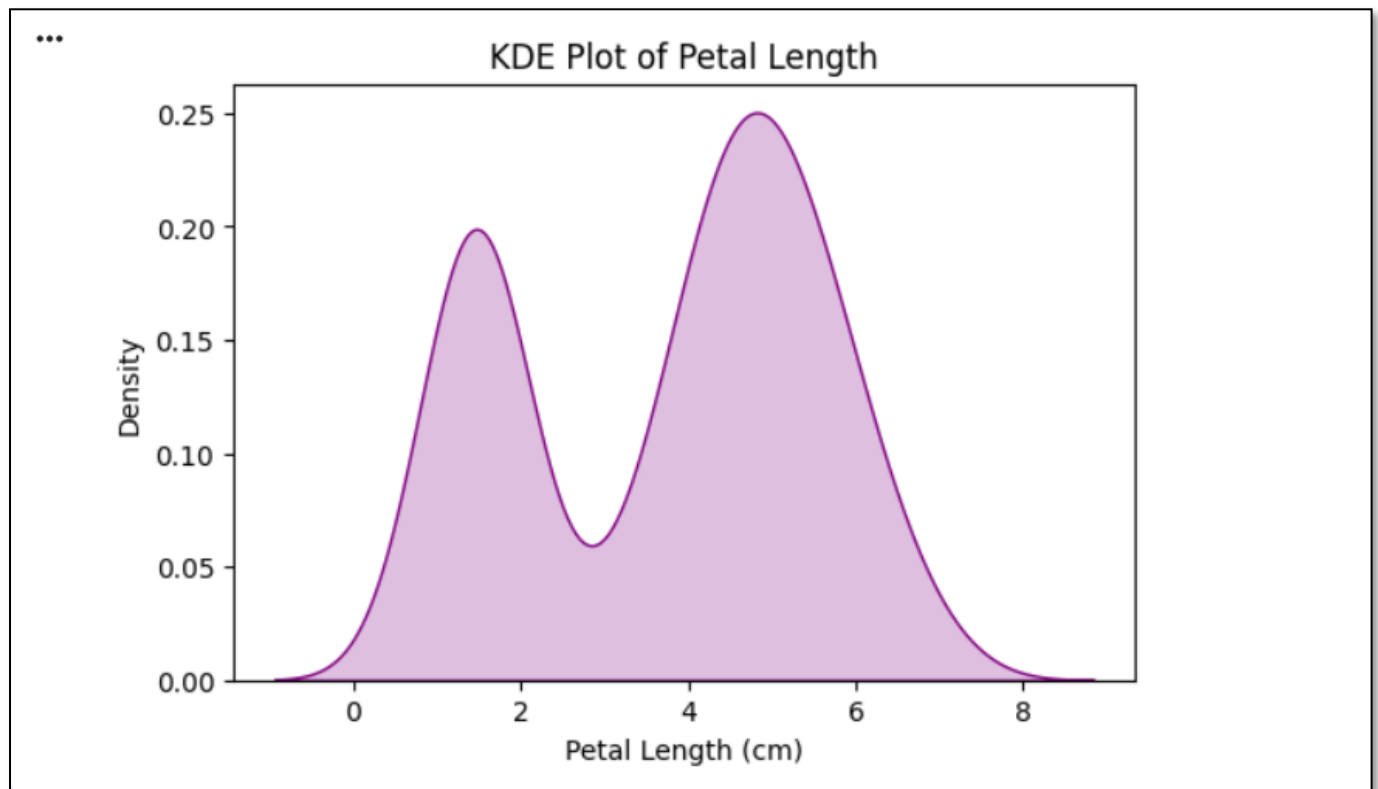
Output:**Observation:**

- The KDE plot shows the density distribution of sepal length.
- The three species have overlapping distributions.
- Iris-setosa tends to have lower sepal lengths than Virginica.
- The overlap indicates that sepal length alone cannot clearly separate all species.

Code:

```
plt.figure(figsize=(6,4))
sns.kdeplot(data=df, x='PetalLengthCm', fill=True, color='purple')

plt.title('KDE Plot of Petal Length')
plt.xlabel('Petal Length (cm)')
plt.ylabel('Density')
plt.show()
```

Output:**Observation:**

- The KDE plot shows the overall distribution of Petal Length.
- The curve indicates where most petal length values are concentrated.
- The distribution is smooth and continuous.
- The plot helps identify the shape and spread of the data.

Post Experimental Exercise:

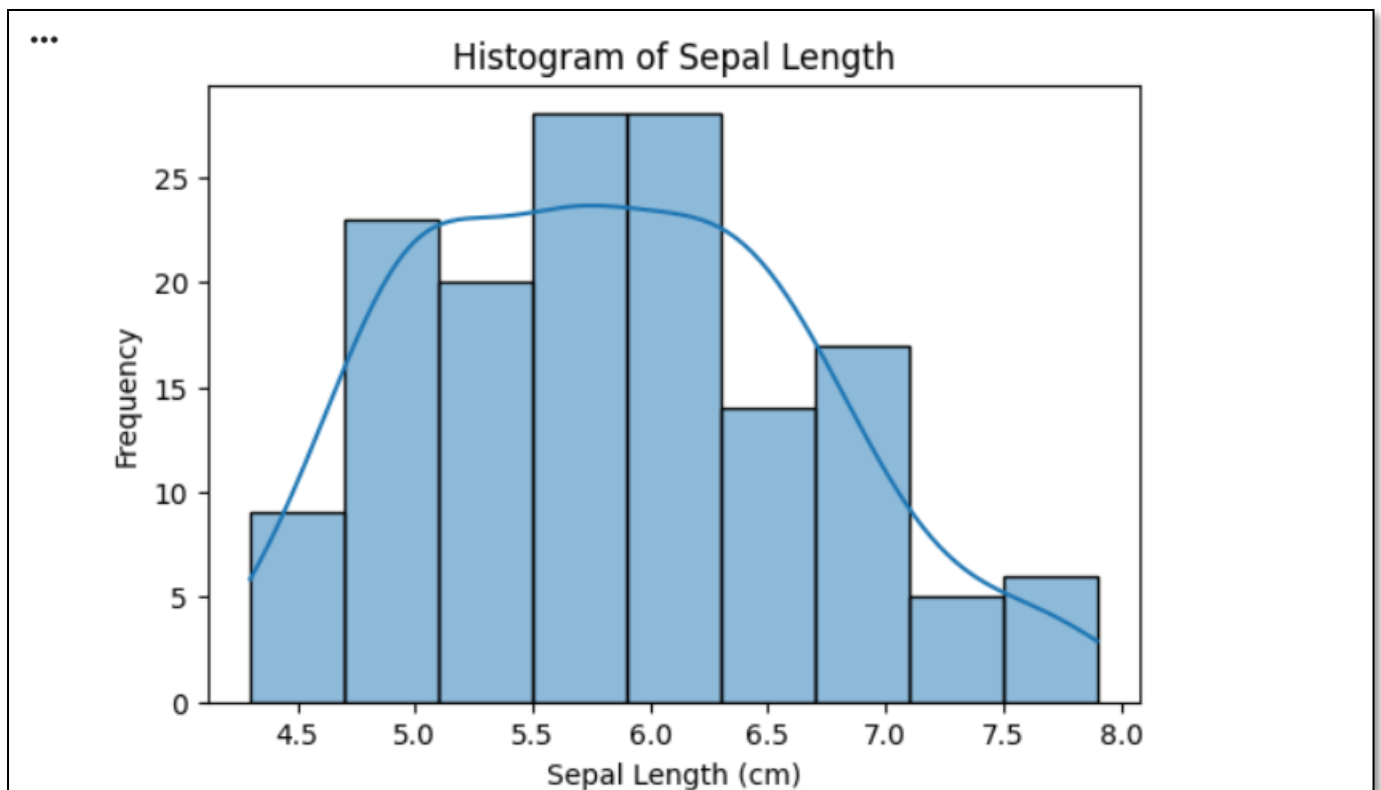
1. A dataset contains a numerical attribute with an unusual distribution. Which visualization technique would you choose to understand its distribution and why?

Code:

```
plt.figure(figsize=(6,4))
sns.histplot(data=df, x='SepalLengthCm', kde=True)
```

```
plt.title('Histogram of Sepal Length')
plt.xlabel('Sepal Length (cm)')
plt.ylabel('Frequency')
plt.show()
```

Output:



Observation:

- Most Sepal Length values are between 5 cm and 7 cm.
- The values are concentrated around the center.
- The KDE curve clearly shows the overall shape of the distribution.
- There are no sudden gaps in the data.

Conclusion:

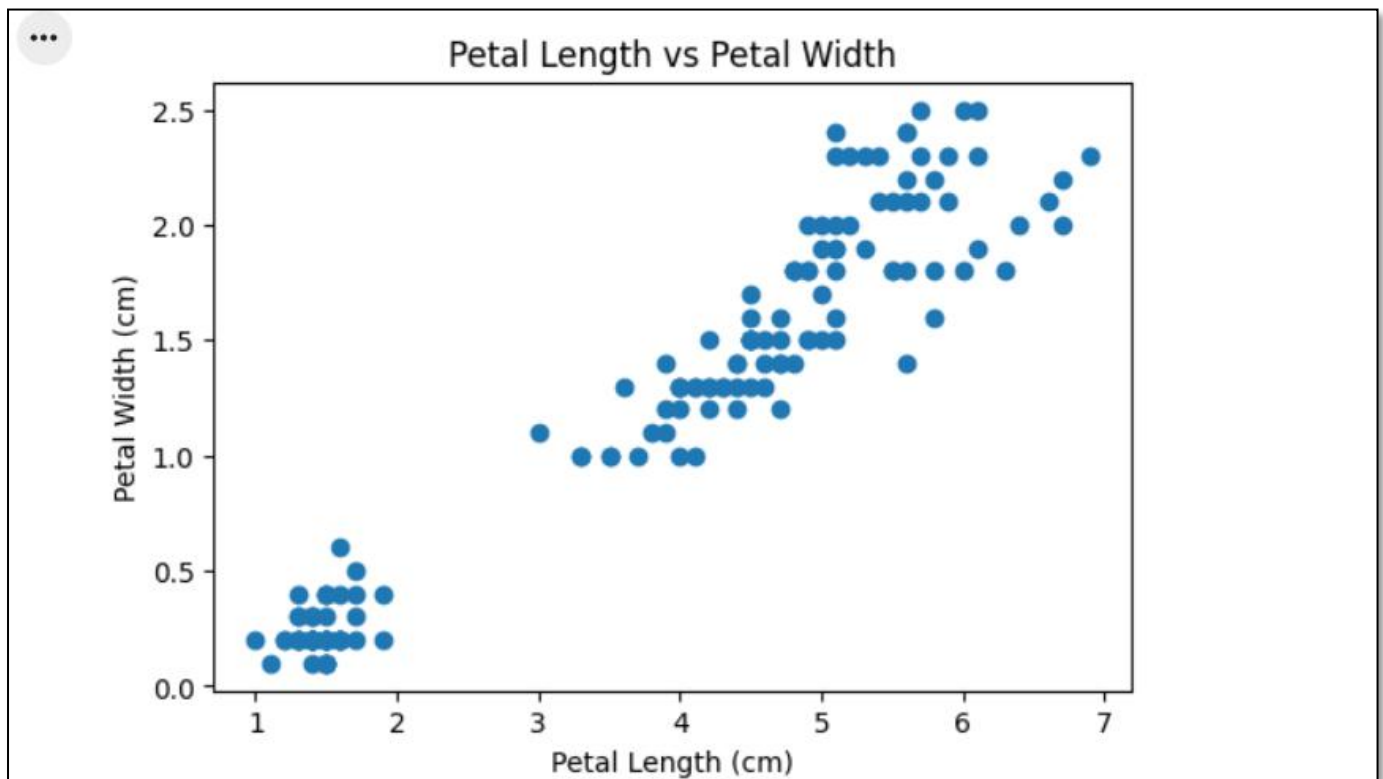
A **Histogram** is the most suitable visualization for understanding the distribution of a numerical attribute because it shows the frequency of values. Using **kde=True** also provides a smooth density curve, making it easier to understand the overall shape of the distribution.

2. Compare the relationship between two numerical attributes using a scatter plot. State whether the relationship appears positive, negative or weak.

Code:

```
plt.figure(figsize=(6,4))
sns.histplot(data=df, x='SepalLengthCm', kde=True)
```

```
plt.title('Histogram of Sepal Length')
plt.xlabel('Sepal Length (cm)')
plt.ylabel('Frequency')
plt.show()
```

Output:**Observation:**

- The points move upward from left to right.
- As Petal Length increases, Petal Width also increases.
- The points are closely clustered with very little scatter.

Conclusion:

The scatter plot shows a **strong positive relationship** between **Petal Length** and **Petal Width**.

3. Download a dataset containing categorical data. Visualize the distribution of a categorical column using a count plot and write your observations.

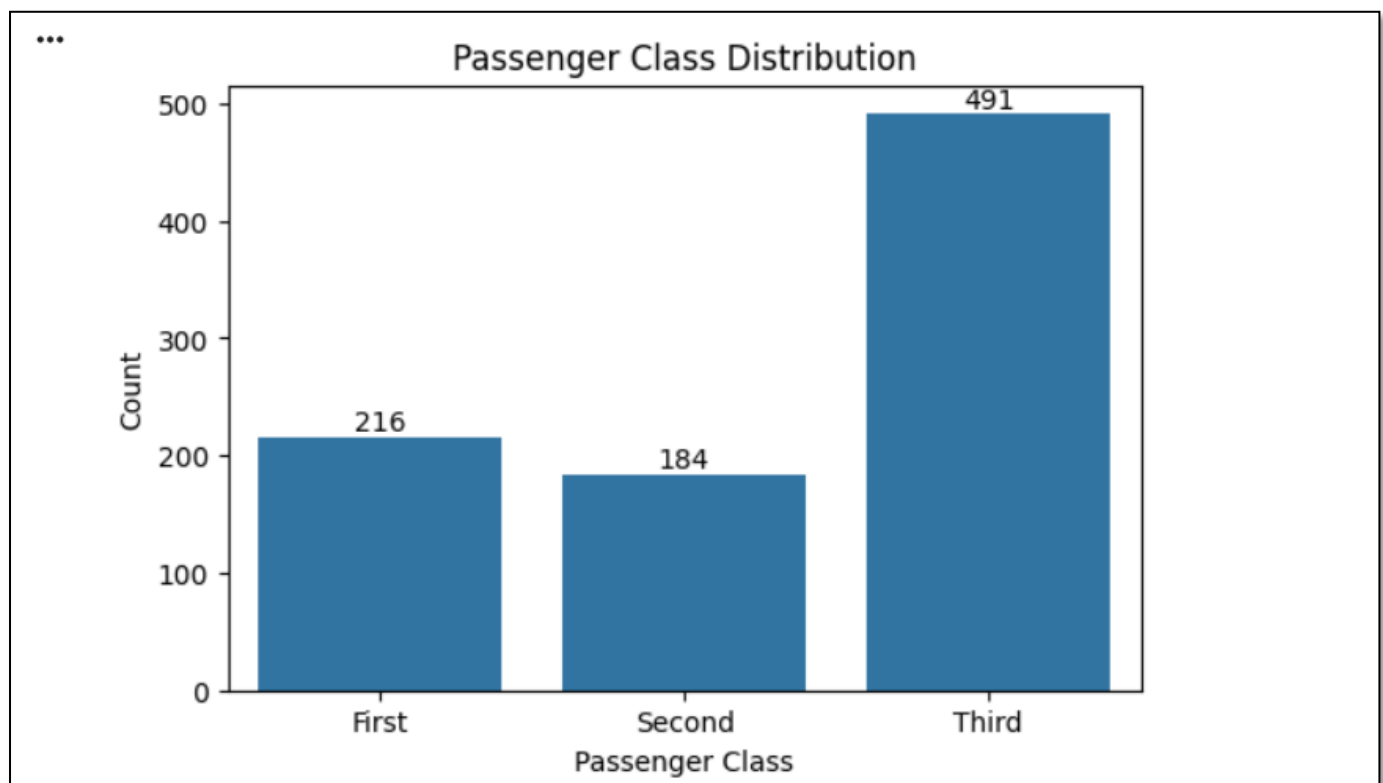
Code:

```
titanic = sns.load_dataset("titanic")
plt.figure(figsize=(6,4))

ax = sns.countplot(data=titanic, x="class")

for container in ax.containers:
    ax.bar_label(container)

plt.title("Passenger Class Distribution")
plt.xlabel("Passenger Class")
plt.ylabel("Count")
plt.show()
```

Output:**Observation:**

- **Third** class has the highest number of passengers.
- **First** class has fewer passengers than Third class.
- The count plot clearly shows the frequency of each passenger class.

Conclusion:

A **Count Plot** is suitable for categorical data because it displays the number of observations in each category, making category-wise comparison simple.